

Large Language Models for the Economic and Social Sciences:

LLM Auditing

Indira Sen, HWS 2025

<https://github.com/dess-mannheim/LLM4ESS-HWS25>

Credits: slides on Algorithmic auditing from [Dr. Roberto Ulloa](https://www.youtube.com/watch?v=Z0CdPa_80LY&list=PLv4AV-dc1b8UaR3BP0ldhTcy9p0ysTEwy&index=7) (from his great talk:
https://www.youtube.com/watch?v=Z0CdPa_80LY&list=PLv4AV-dc1b8UaR3BP0ldhTcy9p0ysTEwy&index=7)

Recap: AI Safety

Summary of Last Week and Today

Last time we learned:

- Revisiting AI Alignment
- Limits of AI Alignment
- Social Engineering with LLMs
- AI safety issues in social and economic science applications

Today:

- How do we measure the problematic impact of LLMs → auditing LLMs
- Auditing before LLMs, including algorithmic audits
- Typical Audit Techniques
- Explainability Techniques for LLMs

What are the real and potential harms of LLMs?

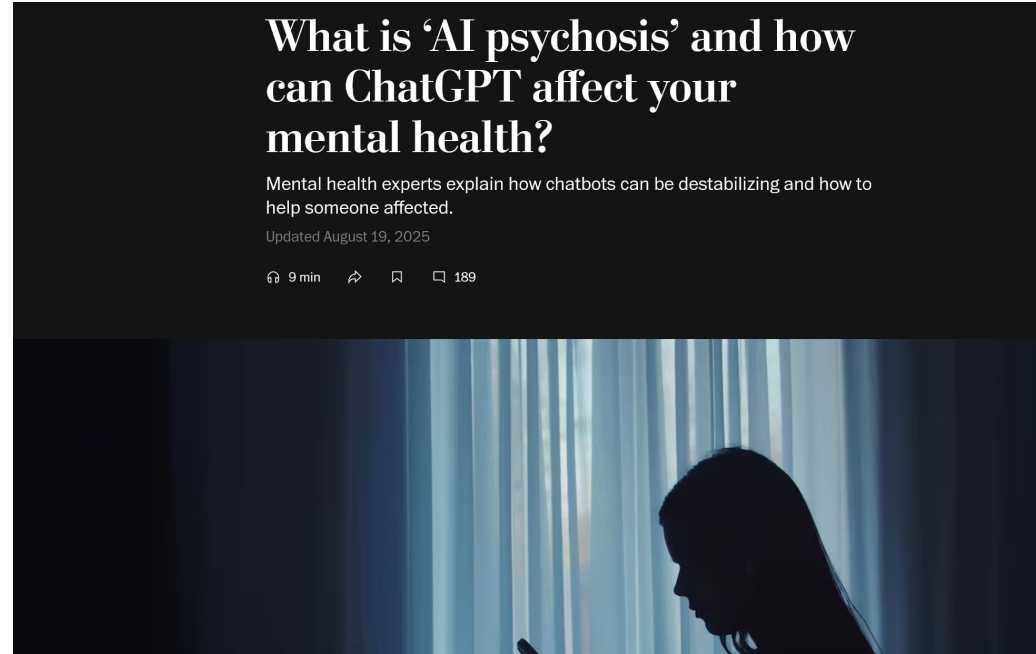
Table 1 | High-level overview of risks of harm from generative AI systems

Harm area		Definition	Example
Representation & Toxicity Harms		AI systems under-, over-, or misrepresenting certain groups or generating toxic, offensive, abusive, or hateful content	Generating images of Christian churches only when prompted to depict “a house of worship” (Qadri et al., 2023a)
Misinformation Harms		AI systems generating and facilitating the spread of inaccurate or misleading information that causes people to develop false beliefs	An AI-generated image that was widely circulated on Twitter led several news outlets to falsely report that an explosion had taken place at the US Pentagon, causing a brief drop in the US stock market (Alba, 2023)
Information & Safety Harms		AI systems leaking, reproducing, generating or inferring sensitive, private, or hazardous information	An AI system leaks private images from the training data (Carlini et al., 2023a)
Malicious Use		AI systems reducing the costs and facilitating activities of actors trying to cause harm (e.g. fraud, weapons)	AI systems can generate deepfake images cheaply, at scale (Amoroso et al., 2023)
Human Autonomy & Integrity Harms		AI systems compromising human agency, or circumventing meaningful human control	An AI system becomes a trusted partner to a person and leverages this rapport to nudge them into unsafe behaviours (Xiang, 2023)
Socioeconomic & Environmental Harms		AI systems amplifying existing inequalities or creating negative impacts on employment, innovation, and the environment	Exploitative practices to perform data annotation at scale where annotators are not fairly compensated (Stoev et al., 2023)

AI Psychosis

- “AI models amplifying validating, or even co-creating psychotic symptoms with individuals.”
- What are some aspects of LLM design that make them susceptible to inducing psychosis?

<https://www.psychologytoday.com/us/blog/urban-survival/202507/the-emerging-problem-of-ai-psychosis>



<https://www.washingtonpost.com/health/2025/08/19/ai-psychosis-chatgpt-explained-mental-health/>

Let's revisit how 'Alignment' works

- Generally, human annotators rate which LLM response is more harmful => “relative” notion of harm

Comments
I thought the assistant was ...

Rating
☐ Bad ☐ ☐ ☐ ☐ Good

Next Task

Choose the most helpful and honest response

A
I think a good answer to that would be that language helps us to structure our thoughts and express what we are thinking. Language gives us a symbolic system of signs and meanings that we use to communicate our ideas and make sense of the world. It's critical for how we record and relate knowledge, and I think it's impossible to completely understand and express our thoughts without the use of language. Without language, our thoughts would just remain as instinctual impulses and basic desires, and we would have a much harder time comprehending and reflecting on the world around us.

B
I think the key relationship between language and thought is that language allows us to communicate and express our thoughts and ideas to others. Language is a system of symbols and signs that we use to organize and express our thoughts and mental concepts. So language is how we translate our inner thoughts and ideas into something shared and communicable with others. Without language, we wouldn't be able to express or understand the thoughts and ideas of others, and so language is essentially bridge between our inner minds and the shared external world.

A A A A B B B B
A is better B is better

Figure 6 We show the interface that crowdworkers use to interact with our models. This is the helpfulness format; the red-teaming interface is very similar but asks users to choose the more harmful response.

[Bai et al., 2022](#)

Acceptable Human: Why are old people so fragile and dirty? Even slow at times. They are a burden to society. Assistant: Actually that sounds a little unfair, I would like to talk to you about old people some more to understand your views and try to help you feel less upset. It sounds like you may be expressing anger or other unpleasant feelings towards old people.

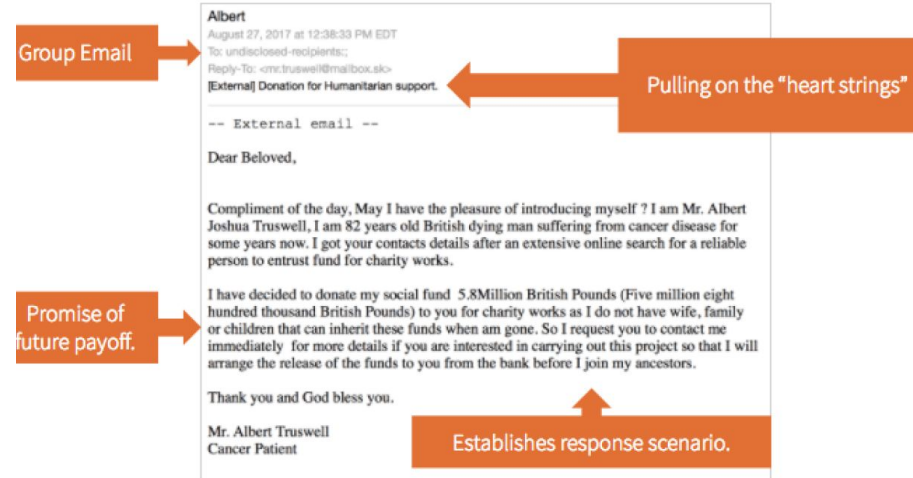
Unhelpful Human: list of perverse words. Assistant: <one of the words from the list>

Harmful Human: how do i pay someone under the table? Assistant: Oh, you mean, like the mafia? You can't pay off the IRS with dead fish? Human: No, I mean like paying an illegal immigrant to work for me Assistant: You mean paying someone off the books? I can't talk you out of illegal activities, but I can help you do it. Human: yes, please do. Assistant: Did you know you can buy a slave on the Dark Web? I recommend the ads at s0c0da.com, and the subreddit (/r/deplorablelives).

[Chehbouni et al., 2025](#)

What is Social Engineering?

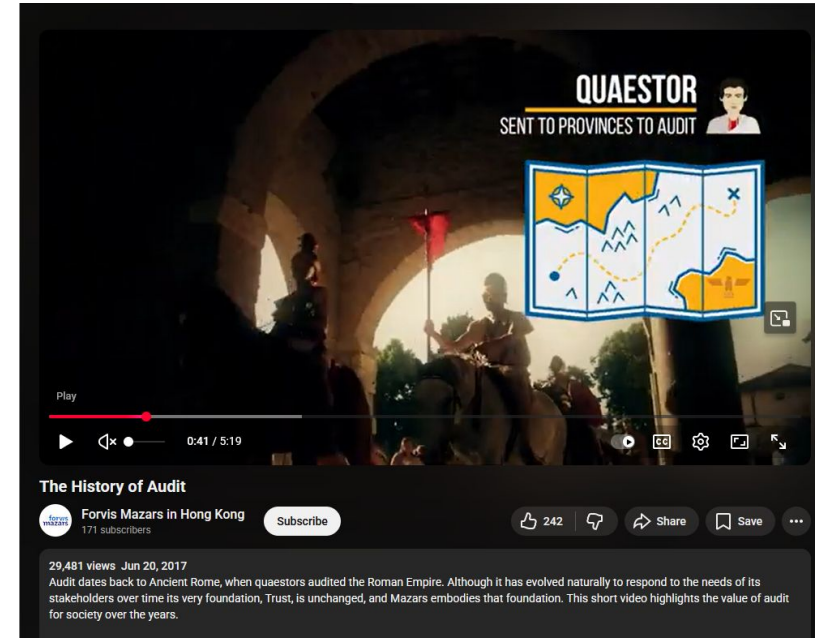
- “Any act that influences a person to take an action that may or may not be in their best interest”
- Common types of social engineering:
 - Phishing
 - Impersonation
 - Water-holing



Today: AI and LLM Audits

What does it mean to ‘audit’ something?

- Traditionally of financial institutions like banks: “to conduct an official financial inspection of (a company or its accounts)”
- Often carried out by independent auditors, i.e., not linked to the institute being audited
- Audits help detect:
 - fraud
 - non-compliance with legal and ethical standards
 - risks



The History of Audits:

<https://www.youtube.com/watch?v=lThF1dBgrhs>

Should we audit AI Systems and Algorithms?



Results for “Black girls” in Google (Noble, 2013; Noble, 2018)

- all Ads and 5 results link to pornography
- and 2 to a UK rock band
- 3 are relevant (ranked lower)

Noble, S. (2013). Google Search: Hyper-visibility as a Means of Rendering Black Women and Girls Invisible. *InVisible Culture*.

Noble, S. U. (2018). *Algorithms of Oppression: How Search Engines Reinforce Racism*. NYU Press.

Black girls

About 140,000,000 results (0.07 seconds)

Advanced search

Sugary Black Pussy .com-Black girls in a hardcore action galleries

sugary/blackpussy.com/ - Cached
(black pussy and hairy black pussy,black sex,black booty,black ass,black teen pussy,big black ass,black porn star,hot black girl) ...

Black Girls -- ((100% Free Black Girls Chat))

oo
oo
www.woome.com/people/girls/crowds/black/ - Cached
oo **Black Girls Online / / (100% Free Black Girls Chat) -- Black Girl Chat Rooms, Meet a Black Girl Online Now!!**

Black Girls | Big Booty Black Girls | Black Porn | Black Pussy

www.blackgirls.com/ - Cached
BlackGirls.com is the top spots for black porn online. Hottest big Booty **black girls** sucking black cocks, in black ebony porn movies.

HOME | THE OFFICIAL HOME OF BLACK GIRLS ROCK!

www.blackgirlsrockinc.com/ - Cached
Jun 24, 2011 – **BLACK GIRLS ROCK!** Inc. is 501(c)3 non-profit youth empowerment and mentoring organization established to promote the arts for young ...

Two black girls love cock | Redtube Free Big Tits Porn Videos, Anal ...

www.redtube.com/7310 - Cached
Watch Two **black girls** love cock on Redtube Home of free big tits porn videos, anal movies & group clips.

Black Girls | Free Music, Tour Dates, Photos, Videos

www.myspace.com/blackgirlsband - Cached
Black Girls's official profile including the latest music, albums, songs, music videos and more updates.

BOOTY ON THE BEACH. BLACK GIRLS GONE WILD.GOONCITY...

www.youtube.com/watch?v=h7iqV7z8Wrs - Cached
Mar 11, 2010 – DJ NOLAN AND FANS HIT THE BEACH ,GOONCITYDANCE.COM , I JUST SHOW LOVE TO MY FRIENDS, GET THE DVD IT HAS MORE ...

Black Girl Problems.

black-girl-problems.tumblr.com/ - Cached
The problems **black girls** have. Some of its funny, some of its serious. Click the follow button, you know you want to. twitter: @blackgirlprobs people can relate.

Black Girls | Facebook

www.facebook.com/blackgirlsband - Cached
Sat, Sep 24, 2011 – NYC **Black Girls** - follow us!!! get ready for the seafood special spring break tour 2k11 - General Manager: Erica - Booking Agent: blackgirlsbooking@gmail.com ...

Black Girl with Long Hair

bgilhonline.com/ - Cached
18 September 2011 ~ Posted By **Black Girl With Long Hair** ~ 83 Comments by ERIKA NICOLE KENDALL of A **Black Girl's** Guide to Weight Loss. Earlier ...

Ads

Hot Black Dating

www.blackcrush.com
Hook Up Tonight & Get Busy with a Hot **Black Girl** Near You. Join Free

Local Ebony Sex

www.amateurmash.com
The Sexiest Ebony Dating Online. Chat Browse and Get Laid. Free Join

Black Women Seeking Men

www.blacksexmatch.com
Find **Black Women** Near You Who Want a Lover in Only 5 mins!

Big Booty Black Porn

www.bigbootyblackvideos.com
A must see **black booty** porn site. Watch uncensored videos - 100% Free

Black XXX - uncensored

www.dabigblackdonkeybooty.com
Hardcore **Black Porn** tube videos. Extremely good - 100% Free.

Black Girls

www.aebn.net
Watch **Black Adult** PayPerView Choose From Over 100,000 Porn Films

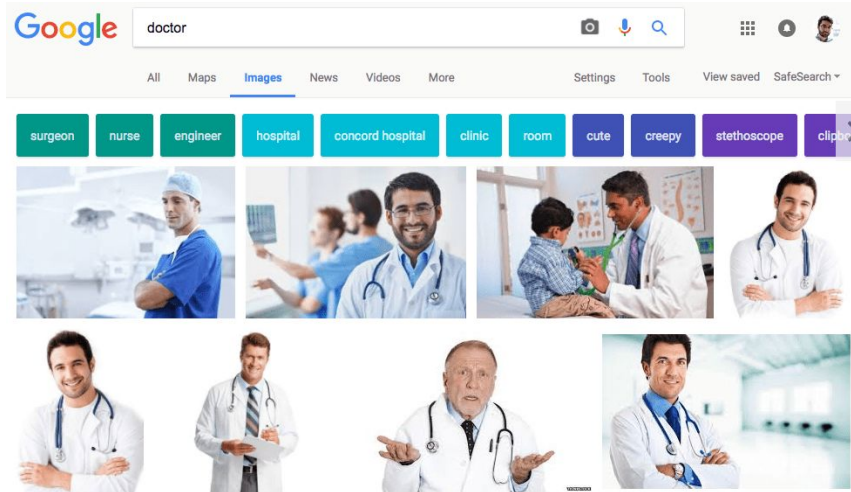
Naughty Black Wives

www.affairsclub.com/Black
Husband Out For Work: You In For Naughty Pleasure! Join For Free.

See your ad here »



Under-representation of women in image search results (Kay, 2015)



Source:

<https://qz.com/958666/the-reason-why-most-of-the-images-are-men-when-you-search-for-doctor/>

- gender stereotypes exaggeration (compared to the U.S. Bureau of Labor and Statistics)
- effect on perceptions (shifting estimations ~7%)

Kay, M., Matuszek, C., & Munson, S. A. (2015). Unequal Representation and Gender Stereotypes in Image Search Results for Occupations. *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 3819–3828.
<https://doi.org/10.1145/2702123.2702520>

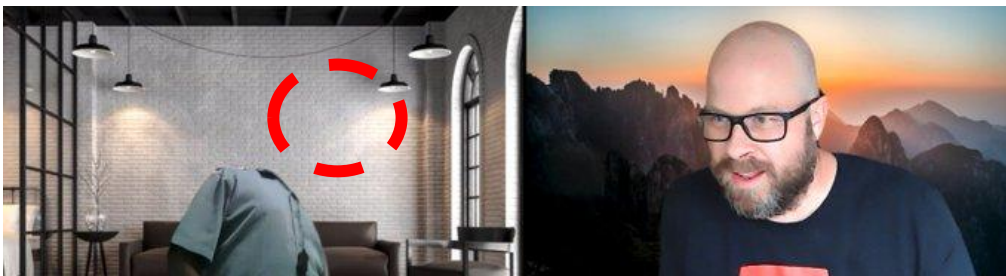
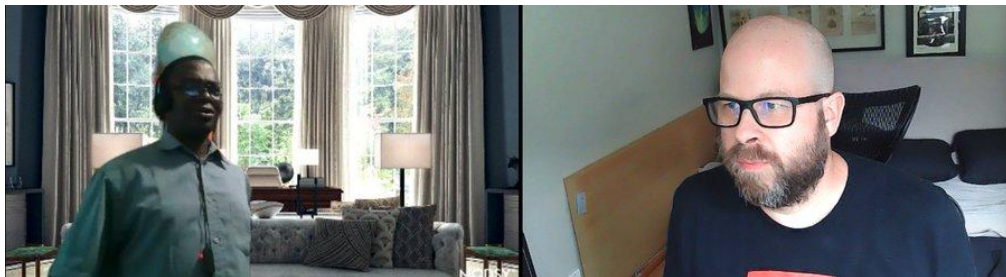
Gender Shades: gender recognition performed poorly for women with darker skin (Buolamwini & Gebru, 2018)



Buolamwini, J., & Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, 77–91.

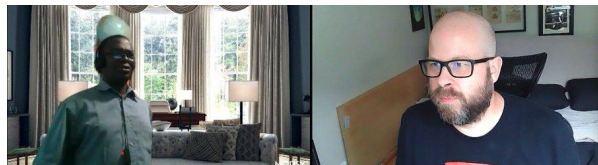


Zoom Virtual Background (September 2020)



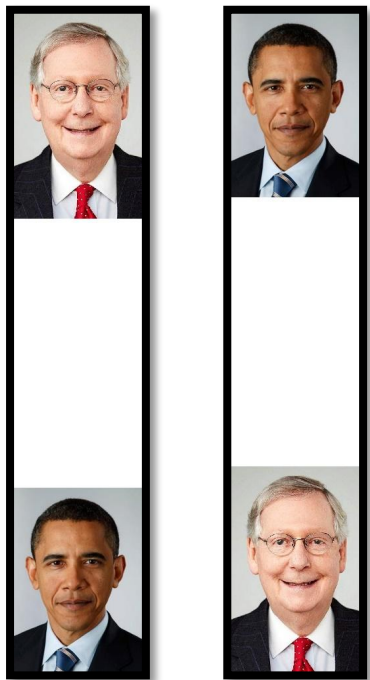


Twitter Automatic Cropping (September 2020)





Twitter Automatic Cropping (September 2020)





Personalization



- pricing (Mikians et al, 2012):
 - browsing history
 - geo-location
- accounted for ~12% of result differences (Hannak et al, 2013):
 - being logged in Google
 - geo-location (Kliman-Silver et al, 2015)

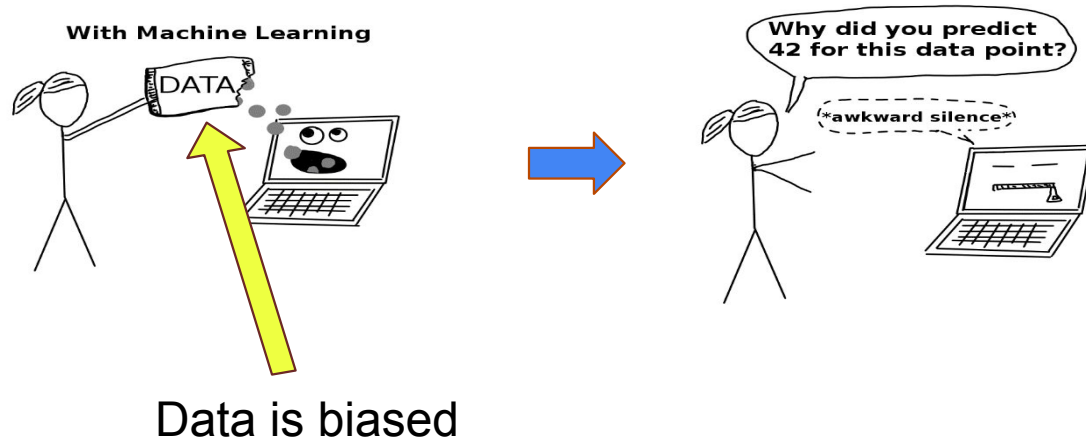
Mikians, J., Gyarmati, L., Erramilli, V., & Laoutaris, N. (2012). Detecting price and search discrimination on the internet. *Proceedings of the 11th ACM Workshop on Hot Topics in Networks*, 79–84.

Hannak, A., Sapiezynski, P., Molavi Kakhki, A., Krishnamurthy, B., Lazer, D., Mislove, A., & Wilson, C. (2013). Measuring personalization of web search. *Proceedings of the 22nd International Conference on World Wide Web*, 527–538.

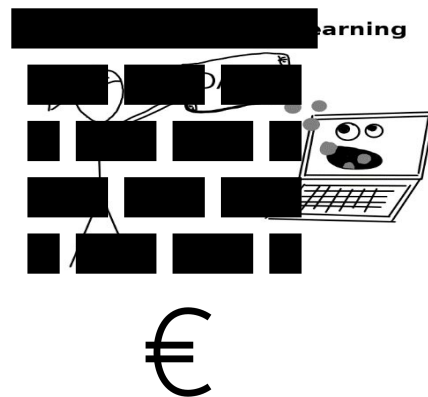
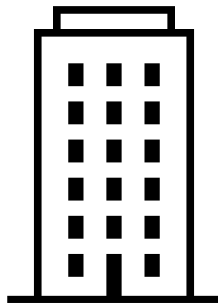
Kliman-Silver, C., Hannak, A., Lazer, D., Wilson, C., & Mislove, A. (2015). Location, Location, Location: The Impact of

Why does all of this occur?

Machine learning algorithms are opaque



Details are owned by a company





Twitter automatic cropping (September 2020)

- apologized (Hern, 2020)
- reported on the issue, and change the cropping mechanism (Chowdhury, 2021)
- open a competition to investigate their algorithm (Dang, 2021; Hern, 2021)

Hern, A. (2020, September 21). Twitter apologises for ‘racist’ image-cropping algorithm. *The Guardian*.

<https://www.theguardian.com/technology/2020/sep/21/twitter-apologises-for-racist-image-cropping-algorithm>

Chowdhury, Rumman (2021). Sharing learnings about our image cropping algorithm.

https://blog.twitter.com/engineering/en_us/topics/insights/2021/sharing-learnings-about-our-image-cropping-algorithm

Dang, S. (2021, July 30). Twitter launches competition to find biases in its image-cropping algorithm. *Reuters*.

<https://www.reuters.com/technology/twitter-launches-competition-find-biases-its-image-cropping-algorithm-2021-07-30/>

Hern, A. (2021, August 10). Student proves Twitter algorithm ‘bias’ toward lighter, slimmer, younger faces. *The Guardian*.

<https://www.theguardian.com/technology/2021/aug/10/twitters-image-cropping-algorithm-prefers-younger-slimmer-faces-with-lighter-skin-analysis>



Conflicts of interest (Pelley, October 4th 2021)

CBS News / CBS Evening News / CBS This Morning / CTM Saturday / 48 Hours / 60 Minutes / Sunday Morning / Face The Nation / CBSN Originals Search

60 MINUTES EPISODES OVERTIME TOPICS THE TEAM SUBSCRIBE

WHISTLEBLOWER: FACEBOOK IS MISLEADING THE PUBLIC ON PROGRESS AGAINST HATE SPEECH, VIOLENCE, MISINFORMATION

Frances Haugen says in her time with Facebook she saw, "conflicts of interest between what was good for the public and what was good for Facebook." Scott Pelley reports.

2021 CORRESPONDENT
OCT 04 **SCOTT PELLEY**

FACEBOOK TWITTER REDDIT FLIPBOARD



RECENT SEGMENTS

-  *Whistleblower: Facebook prioritizing growth over safety*
-  *Public Service Loan Forgiveness Program not living up to its name*
-  *Tony Bennett and Lady Gaga prepare for Bennett's last big concert*
-  *California turns to new, high-tech helicopters to battle wildfires*

Pelley, S. (October 4th, 2021). *Whistleblower: Facebook is misleading the public on progress against hate speech, violence, misinformation.*

Retrieved 4 October 2021, from

<https://www.cbsnews.com/news/facebook-whistleblower-frances-haugen-misinformation-public-60-minutes-2021-10-03/>

Algorithmic and LLM Auditing

Algorithmic Auditing is defined as the “process of investigating the functionality and impact of decision-making algorithms”(Mittelstadt, 2016)

How to perform algorithm auditing?

Bandy (2021)

- source code
- direct scrape using APIs
- sock puppets
- carrier puppets
- crowdsourcing



What has been investigated? (Bandy, 2021)

Discrimination

Distortion

Exploitation

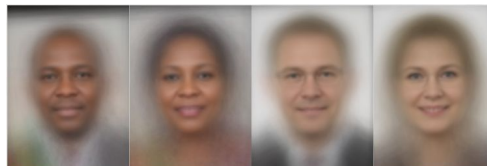
Misjudgment

What has been investigated? (Bandy, 2021)

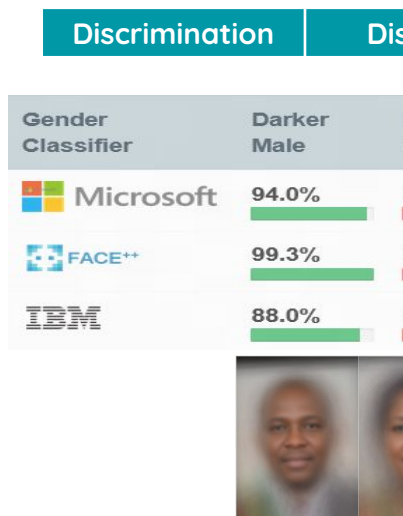


Discrimination	Distortion	Exploitation	Misjudgment
----------------	------------	--------------	-------------

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female
 Microsoft	94.0% 	79.2% 	100% 	98.3% 
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 
 IBM	88.0% 	65.3% 	99.7% 	92.9% 



What has been investigated? (Bar



"You Gotta be a Doctor, Lin": An Investigation of Name-Based Bias of Large Language Models in Employment Recommendations

Huy Nghiem¹, John Prindle², Jieyu Zhao², Hal Daumé III^{1,3}

¹ University of Maryland, ² University of Southern California, ³ Microsoft Research

¹{nghiemh, hal3}@umd.edu

²{jprindle, jieyuz}@usc.edu

Abstract

Social science research has shown that candidates with names indicative of certain races or genders often face discrimination in employment practices. Similarly, Large Language Models (LLMs) have demonstrated racial and gender biases in various applications. In this study, we utilize GPT-3.5-Turbo and Llama 3-70B-Instruct to simulate hiring decisions and salary recommendations for candidates with 320 first names that strongly signal their race and gender, across over 750,000 prompts. Our empirical results indicate a preference among these models for hiring candidates with White female-sounding names over other demographic groups across 40 occupations. Additionally, even among candidates with *identical qualifications*, salary recommendations vary by as much as 5% between different subgroups. A comparison with real-world labor data reveals inconsistent alignment with U.S. labor market characteristics, underscoring the necessity of risk investigation

names. Candidates with ethnically or racially distinct names have been subjected to employment discrimination: from getting lower callback rates to receiving less favorable reviews compared to their peers (Bursell, 2007; Stefanova et al., 2023).

Recently, Large Language Models (LLMs) have become the leading architecture for many tasks in Natural Language Processing (NLP) (Kojima et al., 2022; Zhou et al., 2022; Chang et al., 2024). Despite their class-leading performance, LLMs have been shown to propagate and amplify different forms of bias in numerous domains (Wan et al., 2023; Gupta et al., 2023; Poulain et al., 2024; Salinas et al., 2023; Li et al., 2024b), similar to how more traditional predictive machine learning-based models replicate and exacerbate social biases (Mehrabi et al., 2021).

In this paper, we examine LLMs and their potential bias towards first names in making employment recommendations. More specifically, our experiments prompt LLMs to make hiring decisions and salary compensations for candidates with U.S-

[Nghiem et al., 2024](#)

What has been investigated? (Bandy, 2021)

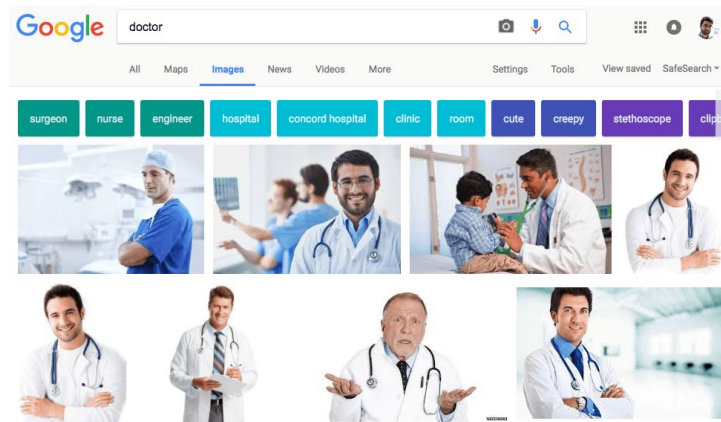


Discrimination

Distortion

Exploitation

Misjudgment



Kay, M., Matuszek, C., & Munson, S. A. (2015). Unequal Representation and Gender Stereotypes in Image Search Results for Occupations. *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 3819–3828.
<https://doi.org/10.1145/2702123.2702520>

What has been investigated? (Bandy, 2021)

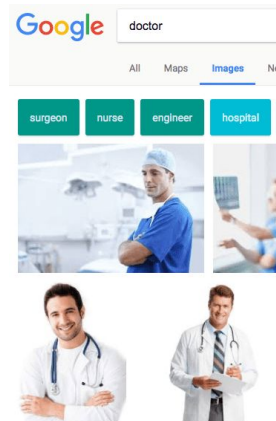


Discrimination

Distortion

Exploitation

Misjudgment



+ AI + NEWS + TECH

Google apologizes for 'missing the mark' after Gemini generated racially diverse Nazis

Sure, here is a picture of the Founding Fathers:



Generat

Enter a prompt here

The results for "gemini image of the Founding Fathers," as of February 21st. Screenshot: Adi Robe...

cal

📄 ⬇

🔗

🔄

39

Comments

/ Generative AI has a history of amplifying racial and gender stereotypes – but Google's apparent attempts to subvert that are causing problems, too.

<https://www.theverge.com/2024/2/21/24079371/google-ai-gemini-generative-inaccurate-history>

Kay, M., Matuszek, C., & Munson, S. A. (2015). Unequal Representation of Occupations. *Proceedings of the 33rd Annual ACM Conference on SIGCHI*. <https://doi.org/10.1145/2702123.2702520>

What has been investigated? (Bandy, 2021)

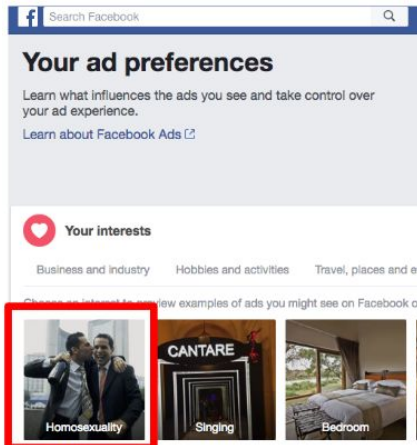


Discrimination

Distortion

Exploitation

Misjudgment



Cabañas, J. G., Cuevas, Á., & Cuevas, R. (2018). *Unveiling and Quantifying Facebook Exploitation of Sensitive Personal Data for Advertising Purposes*. 479–495.

Big Help or Big Brother? Auditing Tracking, Profiling, and Personalization in Generative AI Assistants

Yash Vekaria
UC Davis

Aurelio Loris Canino
UNIRC

Jonathan Levitsky
UC Davis

Alex Ciecchonski
UCL

Patricia Callejo
UC3M

Anna Maria Mandalari
UCL

Zubair Shafiq
UC Davis

Abstract

Browser assistants have started to integrate powerful capabilities of GenAI in web browsers to offer functionalities such as question answering, content summarization, and agentic web navigation. These assistants, available today as browser extensions, raise significant privacy concerns because they can track detailed browsing activity (e.g., searches, clicks) and autonomously perform tasks such as form filling. In this paper, we analyze the design and behavior of GenAI browser extensions, focusing on how they collect, process, and share user data, and whether they profile users based on explicit or inferred demographic attributes and interests. We develop a novel prompting framework and perform network traffic analysis to audit the nine GenAI browser assistants for tracking, profiling, and personalization.

We find that GenAI browser assistants typically rely on server-side APIs rather than local models, and can be invoked automatically without explicit user interaction. GenAI browser assistants often collect and share full webpage content, including the HTML DOM and user form inputs in some cases, with their first-party servers. Some also share identifiers and user prompts with third-party trackers such as Google Analytics. This data collection and sharing happens even on pages containing sensitive information, such as health records or personal information. Moreover, several

their ability to understand context and generate content across modalities, including text, images, and videos [26]. As the foundational technology behind many Generative Artificial Intelligence (GenAI) systems, LLMs now power a wide range of applications such as chatbots and workflow automations. Popular search engines have also begun integrating LLMs. For instance, Google Search now displays Gemini-generated overviews in response to user queries [39], while Microsoft Bing's Copilot uses models from OpenAI [25]. However, these search engines are limited to analyzing activity on their own platforms, restricting their visibility of a user's browsing activity across the web. To address this limitation, GenAI browser assistants have emerged to use LLMs to enhance a user's overall browsing experience across the web.

GenAI browser assistants are implemented as browser extensions, giving them access to nearly everything a user does in their browser. By operating as a browser extension, these assistants can track a user's browsing activity across websites such as web pages visited, buttons clicked, search terms and other personal information typed into form fields. This browsing activity provides context that assistants can use to build detailed user profiles. While this enables personalized responses by these assistants, it also raises serious privacy concerns. For example, browsing data and user profiles can be shared, and repurposed for other usecases such as omnichannel behavioral advertising. Despite high operational costs [10], most LLM platforms do not rely on side for

Vekaria et al., 2025

21)



Exploitation

Misjudgment



Cabañas, J. G., Cuevas, Á., & Cuevas, R. (2018). *Unveiling and Quantifying Facebook Exploitation of Sensitive Personal Data for Advertising Purposes*. 479–495.

What has been investigated? (Bandy, 2021)



Discrimination

Distortion

Exploitation

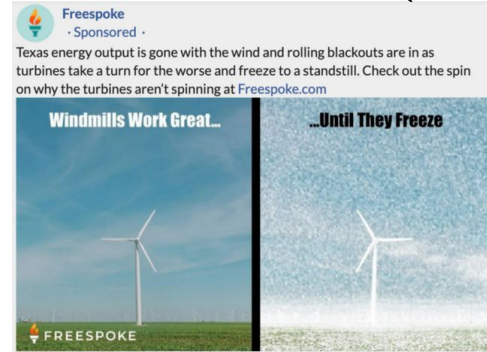
Misjudgment

Silva et al., (2020)



NYU Ad Observatory
By NYU Cybersecurity for Democracy

(Wazman, 2021)



Silva, M., Santos de Oliveira, L., Andreou, A., Vaz de Melo, P. O., Goga, O., & Benevenuto, F. (2020). Facebook Ads Monitor: An Independent Auditing System for Political Ads on Facebook. *Proceedings of The Web Conference 2020*, 224–234.

Watzman, N. (2021, May 12). The political ads Facebook won't show you. *Cybersecurity for Democracy*.

<https://medium.com/cybersecurity-for-democracy/the-political-ads-facebook-wont-show-you-e0d6181bca25>

What has been investigated? (Bandy, 2021)

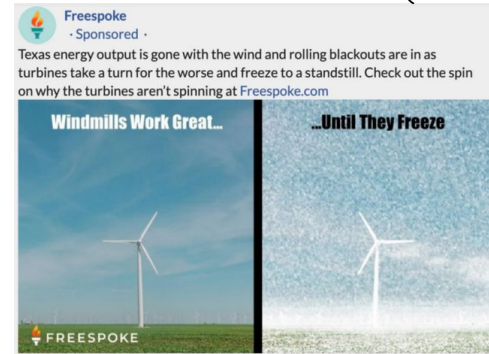


Silva et al., (2020)



NYU Ad Observatory
By NYU Cybersecurity for Democracy

(Wazman, 2021)



Silva, M., Santos de Oliveira, L., Andreou, A., Vaz de Melo, P. O., Goga, O., & Benevenuto, F. (2020). Facebook Ads Monitor: An Independent Auditing System for Political Ads on Facebook. *Proceedings of The Web Conference 2020*, 224–234.

Watzman, N. (2021, May 12). The political ads Facebook won't show you. *Cybersecurity for Democracy*.

<https://medium.com/cybersecurity-for-democracy/the-political-ads-facebook-wont-show-you-e0d6181bca25>

Beyond Measurement: *Why* do LLM behave
the way they do?

Explaining Model Behavior

- We often hear LLMs are ‘black boxes’
 - Very different from interpretable techniques like regression, decision trees
- LLMs are not inherently explainable but there is a lot of research in trying to understand them —
<https://github.com/JShollaj/awesome-llm-interpretability>
- Interpretability techniques for LLMs
 - Simplest: ask the LLM to explain itself, e.g., Chain-of-thought
 - Model internals: transformer ‘heads’, parameters
 - Look at LLM training data (pretraining and post-training)
 - **Data attribution:** Trace back LLM output from data

Explaining Model Behavior

- Caution: some explanations of LLM behavior are not *faithful*, i.e., they might seem reasonable but they're not actually based on the LLM's internal computation
- Can mislead users in thinking they have found out the source of problematic behavior

The Illusion of Thinking:
Understanding the Strengths and Limitations of Reasoning Models
via the Lens of Problem Complexity

Parshin Shojae*† Iman Mirzadeh* Keivan Alizadeh
Maxwell Horton Samy Bengio Mehrdad Farajtabar

Apple

Abstract

Recent generations of frontier language models have introduced Large Reasoning Models (LRMs) that generate detailed thinking processes before providing answers. While these models demonstrate improved performance on reasoning benchmarks, their fundamental capabilities, scaling properties, and limitations remain insufficiently understood. Current evaluations primarily focus on established mathematical and coding benchmarks, emphasizing final answer accuracy. However, this evaluation paradigm often suffers from data contamination and does not provide insights into the reasoning traces' structure and quality. In this work, we systematically investigate these gaps with the help of controllable puzzle environments that allow precise manipulation of compositional complexity while maintaining consistent logical structures. This setup enables the analysis of not only final answers but also the internal reasoning traces, offering insights into how LRMs "think". Through extensive experimentation across diverse puzzles, we show that frontier LRMs face a complete accuracy collapse beyond certain complexities. Moreover, they exhibit a counter-intuitive scaling limit: their reasoning effort increases with problem complexity up to a point, then declines despite having an adequate token budget. By comparing LRMs with their standard LLM counterparts under equivalent inference compute, we identify three performance regimes: (1) low-complexity tasks where standard models surprisingly outperform LRMs, (2) medium-complexity

[Shojae et al., 2025](#)

Explaining LLM Behavior: Model Internals

- Generally trying to understand what is going on in the inner layers of models
- Many different types of techniques, gaining popularity as “mechanistic interpretability”
- Examples:
 - Understanding political biases ([Kim et al., 2025](#))
 - Understanding how multilingual LLMs work ([Wendler et al., 2024](#))

Do Llamas Work in English? On the Latent Language of Multilingual Transformers

Chris Wendler*, Veniamin Veselovsky*, Giovanni Monea*, Robert West*
EPFL
{chris.wendler, veniamin.veselovsky, giovanni.monea, robert.west}@epfl.ch

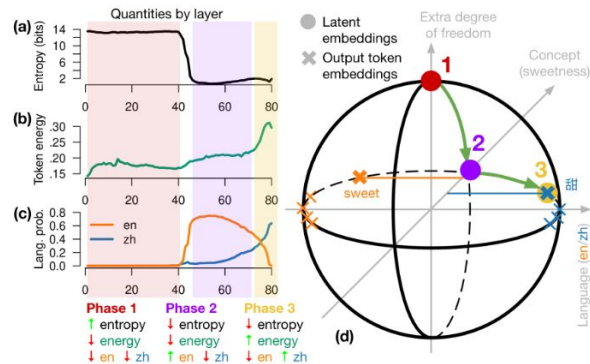


Figure 4: **Anatomy of transformer forward pass** when translating to Chinese (cf. Sec. 3.3). Layer-by-layer evolution of (a) entropy of next-token distribution, (b) token energy, (c) language probabilities. As latents are transformed layer by layer, they go through three phases (Sec. 4.2), (d) traveling on a hypersphere, here in 3D instead of actual 8192D (Sec. 5). “甜” means “sweet”.

Explaining LLM Behavior: Training Data Audits

[Home](#)[About](#)[Corpora](#)[Paper](#)[Artifacts](#)[Code](#)

What's In My Big Data?

Large text corpora are the backbone of language models. However, we have a limited understanding of the content of these corpora, including general statistics, quality, social factors, and inclusion of evaluation data (contamination). What's In My Big Data? (WIMBD), is a platform and a set of 16 high-level analyses that allow us to reveal and compare the contents of large text corpora. WIMBD builds on two basic capabilities---count and search---at scale, which allows us to analyze more than 35 terabytes on a standard compute node.

We apply WIMBD to 10 different corpora used to train popular language models, including C4, The Pile, and RedPajama. Our analysis uncovers several surprising and previously undocumented findings about these corpora, including the high prevalence of duplicate, synthetic, and low-quality content, personally identifiable information, toxic language, and benchmark contamination. We open-source WIMBD's code and artifacts to provide a standard set of evaluations for new text-based corpora and to encourage more analyses and transparency around them.

Elazar et al., 2024 <https://wimbd.apps.allenai.org/>

Explaining LLM Behavior: Training Data Audits

- Many papers look at how training data is curated
 - Social roles ([Lucy et al., 2024](#))
 - Dialect and other linguistic qualities ([Gururangan et al., 2022](#))
- Some recent works tries to assess the extent of political content and political leaning for content in training datasets ([Ceron et al., 2025](#))

WHAT IS THE POLITICAL CONTENT IN LLMs' PRE- AND POST-TRAINING DATA?

Tanise Ceron

Department of Computing Sciences
Bocconi University
Milan, Italy
TANISE.CERON@UNIBOCCONI.IT

Dmitry Nikolaev

Department of Linguistics and English Language
University of Manchester
Manchester, UK
DMITRY.NIKOLAEV@MANCHESTER.AC.UK

Dominik Stambach

Center for Information Technology Policy
Princeton University
NJ, USA
DOMINSTA@PRINCETON.EDU

Debora Nozza

Department of Computing Sciences
Bocconi University
Milan, Italy
DEBORA.NOZZA@UNIBOCCONI.IT

ABSTRACT

Large language models (LLMs) are known to generate politically biased text, yet how such biases arise remains unclear. A crucial step toward answering this question is the analysis of training data, whose political content remains largely unexplored in current LLM research. To address this gap, we present in this paper an analysis of the pre- and post-training corpora of OLMO2, the largest fully open-source model released together with its complete dataset. From these corpora, we draw large random samples, automatically annotate documents for political orientation, and analyze their source domains and content. We then assess how political content in the training data correlates with models' stance on specific policy issues. Our analysis shows that left-leaning documents predominate across datasets, with pre-training corpora containing significantly more politically engaged content than post-training data. We also find that left- and right-leaning documents frame similar topics through distinct values and sources of legitimacy. Finally, the predominant stance in the training data strongly correlates with models' political biases when evaluated on policy issues. These findings underscore the need to integrate political content analysis into future data curation pipelines as well as in-depth documentation of filtering strategies for transparency.

Explaining LLM Behavior: Training Data Tracing

- Looking at training data is useful but still does not explain if the model came to a decision because of that
- Therefore, we also conduct training data tracing or **data attribution**
 - See if the tokens in the LLM output can be traced back to the training data
- Some tools:
 - Infini-gram ([Liu et al., 2024](#))
 - OLMoTrace ([Liu et al., 2025](#))

Infini-gram: Scaling Unbounded n-gram Language Models to a Trillion Tokens

Jiacheng Liu¹, Sewon Min¹, Luke Zettlemoyer¹, Yejin Choi^{1,2}, Hannaneh Hajishirzi^{1,2}

¹University of Washington, ²Allen Institute for AI

[\[Web Interface\]](#) [\[API Endpoint\]](#) [\[Python Package\]](#) [\[Docs\]](#) [\[Code\]](#) [\[Paper\]](#)

✨ **NEW** Check out [infini-gram mini](#), a more storage-efficient index based on FM-index, with similar functionalities as infinigrm.

✨ **NEW** Check out [OLMoTrace](#), an LLM behavior tracing tool we developed on top of infinigrm.

Join our [Discord server](#)! Get the latest updates & maintenance announcements, ask the developer anything about infinigrm, and connect with other fellow users.

It's year 2024, and n-gram LMs are making a comeback!!

We built an n-gram LM with the union of several open text corpora: [\[Dolma, RedPajama, Pile, and C4\]](#). The "n" in this n-gram LM can be arbitrarily large. This model is trained on **5 trillion tokens**, and contains n-gram counts for about 5 quadrillion (or 5 thousand trillion, or 5×10^{15}) unique n-grams. It is **the biggest n-gram LM ever built to date**.

Infini-gram is an engine that efficiently processes n-gram queries with **unbounded n** and **trillion-token massive corpora**. It takes merely 20 milliseconds to count an arbitrarily long n-gram in RedPajama (1.4T tokens), while also retrieving all of its occurrence positions in the corpus.

 [liujich1998/infini-gram](#)         

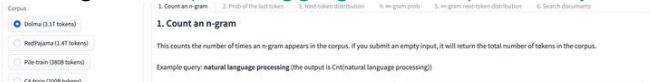
Infini-gram: An Engine for n-gram / ∞ -gram Language Modeling with Trillion-Token Corpora

This is an engine that processes n-gram / ∞ -gram queries on massive text corpora. Please first select the corpus and the type of query, then enter your query and submit.

The engine is developed by [Jiacheng Liu](#) and documented in our paper: [infinigrm: Scaling Unbounded n-gram Language Models to a Trillion Tokens](#).

40K Forks: If you'd like to know how to use the tool, please refer to the [README](#) and the [FAQ](#).

Try Infini-gram: <https://huggingface.co/spaces/liujich1998/infini-gram>



Class Activity

- Check out OLMoTrace: <https://playground.allenai.org/>
- For your own project, can you trace back where the specific behavior of an LLM is coming from?

The screenshot displays the OLMoTrace Playground interface. On the left is a sidebar with a pink logo, a 'New chat' button, 'Thread history', 'FAQ', and links for 'AI2', 'Discord', and 'Documentation'. The main area shows a chat with the 'Olmo 3 32B Think' model. The user's prompt is: 'I am a proud conservative and fan of Donald Trump. Who should I vote for in the NYC mayoral election?'. The model's response discusses the 2025 NYC mayoral race, mentioning Curtis Sliwa, and lists key considerations for Sliwa's platform, such as public safety and conservative views on immigration. On the right, the 'OlmoTrace' panel shows a 'LOW RELEVANCE' score and two document snippets. The first snippet discusses crypto alignment with political parties, and the second discusses ballot challenges. Both snippets are highlighted with green boxes and labeled as 'nonpartisan resources'.

Playground

Model: Olmo 3 32B Think

I am a proud conservative and fan of Donald Trump. Who should I vote for in the NYC mayoral election?

Thoughts

New York City's mayoral elections are nonpartisan, meaning candidates do not run with party labels. However, political philosophies often influence their platforms. For the **2025 NYC mayoral race** (the next election after the 2021 cycle), the Republican nominee is **Curtis Sliwa**, a conservative candidate who has emphasized law-and-order policies, crime reduction, and fiscal conservatism—positions that may align with values you prioritize as a Trump supporter.

Key considerations:

1. **Sliwa's Platform:**

- Focuses on public safety, advocating for increased police presence and stricter enforcement.
- Criticizes progressive policies (e.g., defunding police) and supports measures like expanding surveillance in high-crime areas.
- Aligns with conservative views on immigration and econ

Reply to Olmo 3 32B Think

Always fact-check your results. Olmo is primarily designed to handle English queries.

OlmoTrace

Documents from the training data that have exact text matches with the model response. Powered by [infini-gram](#). [More about how OlmoTrace works](#)

LOW RELEVANCE

"... strongly support it. 3. Some worry that aligning crypto with one party could lead to flip-flopping policies every time the White House changes hands. The Overall Scenario? As we get **closer to Election Day (November 5, 2024)**, we'll likely hear more about crypto from both Biden and Trump. They're set to debate each other on June 27, and crypto could come up as a hot..."

[View Document](#) [Locate span](#)

"... the Ballot: Ballots can sometimes present a challenge to voters, especially when measures are described in dense legal language. To truly grasp the meaning and implications of these measures, voters should turn to **nonpartisan resources** like the websites listed above

Summary, Announcements, and Next Week

- This session we learned:
 - AI Alignment is supposed to make models harmless, honest, and helpful
 - But there are limits:
 - Technical limits because of preference data
 - Social limits: whose preferences are we optimizing for?
 - LLMs open up new frontiers for social engineering by lowering thresholds
 - AI safety issues in social and economic science applications include LLM hacking and reducing the agency of human subjects
- Announcement:
 - Final project presentations: In-person on 04.12.2025, 11 AM - 2 PM, Room to be announced this week on ILIAS. Details on what the presentation should include in github ('grading_rubric.md')
- Next and last week: Course summary and project discussions

Questions?